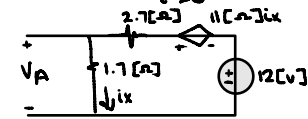


continue Part b)



KCL

$$12[V] = 1.7[\Omega]i_X + 2.7[\Omega]i_X - 11[\Omega]i_X$$

$$i_X = -1.88[A]$$

$$V_A = i_X \cdot 1.7[\Omega] = -3.09[V]$$

$$V_A(t) = (-3.09 + \frac{2.96}{1.7}e^{-t/3.826})[V]$$

from the inverse transform.

$$V_A(t) = 4.00[V] \cos(70[\text{mrad/s}]t - 42.3^\circ)$$

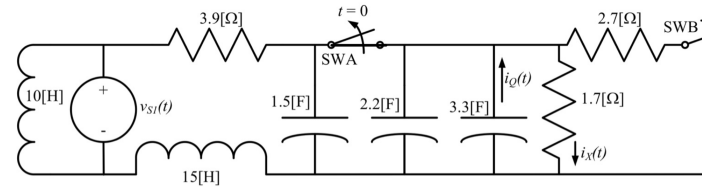
$$V_A(0) = 4.00[V] \cos(-42.3^\circ) = 2.96[V]$$

$$i_X(2cs) = \frac{V_A(2cs)}{1.7[\Omega]} = \frac{(-3.09 + 6.05e^{2/3.826})[V]}{1.7[\Omega]}$$

$$i_X(2cs) = 4.19[A]$$

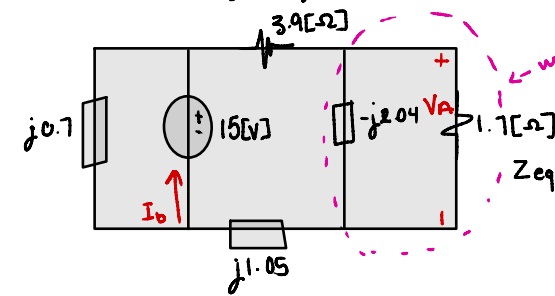
Set up

t < 0



$$C_{eq} = 1.5[F] + 2.2[F] + 3.3[F] = 7.0[F]$$

Convert: $V_{S1}(t) = 15[V] \cos(70[\text{mrad/s}]t)$
 $Z_L = j\omega L \rightarrow j(70e-3)(10[H]) \rightarrow j0.7[\Omega]; j(70e-3)(15[H]) = j1.05[\Omega]$
 $Z_C = \frac{j}{\omega C} \rightarrow \frac{j}{(70e-3)(7[F])} \rightarrow -j2.04[\Omega]$



we can find Zeq from this area for V_A

$$Z_{eq} = 1.7[\Omega] \parallel -j2.04[\Omega]$$

$$\frac{(1.7 \cdot -j2.04)[\Omega]}{1.7[\Omega] - j2.04[\Omega]} = 1.31[\Omega] \angle -39.8^\circ$$

$$V_A = 15[V] \left(\frac{1.31 \angle -39.8^\circ}{3.9 + j1.05 + 1.31 \angle -39.8^\circ} \right) = 4[V] \angle -42.3^\circ$$

I believe we don't include j0.7 cause we are seeing the circuit from V_A's pov.

$$I_0 = \frac{15[V]}{j0.7[\Omega]} + \frac{15[V]}{1.9 + j1.05 + 1.31 \angle -39.8^\circ} = 21.8 \angle -81.9^\circ[A]$$

in time domain: $21.8[A] \cos(70[\text{mrad/s}]t - 81.9^\circ)$

Part A) Find the power delivered by $V_{S1}(t)$ at $t=0$

$$P_{del, V_{S1}} = V_{S1}(t) \cdot I_0(t)$$

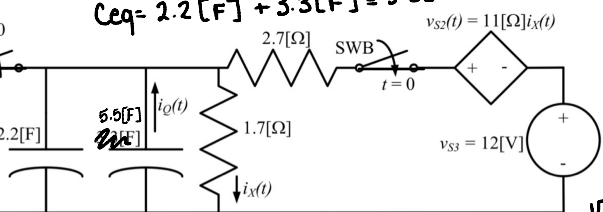
$$(15[V] \angle 0^\circ)(21.8[A] \angle -81.9^\circ)$$

$$= 46.1[W]$$

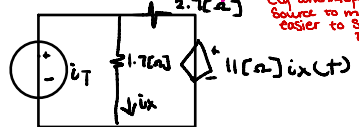
Part b) Find $i_X(2cs)$

t > 0

$$C_{eq} = 2.2[F] + 3.3[F] = 5.5[F]$$



We need to find Req. We take out the Ceq and independent source to make it easier to solve for Req.



$$i_X = \frac{1[V]}{1.7[\Omega]} = 0.588[A]$$

$$i_T = \frac{1[V]}{1.7[\Omega]} + \frac{1[V] - 11[\Omega]i_X}{2.7[\Omega]}$$

$$= -1.438[A]$$

$$R_{eq} = \frac{1[V]}{-1.438[A]} \rightarrow -0.695$$

$$Z = R_{eq} C_{eq} \rightarrow (5.5[F])(-0.695[\Omega]) = -3.825[cs]$$